Portfolio Theory, Risk, and Return

Corporate Finance
Chapter 13

I. Introduction to Portfolios

A **portfolio** is a group of assets held by an investor (e.g., a combination of stocks and bonds). The **portfolio weight** represents the percentage of the total investment value allocated to each asset.

Just as individual assets have expected returns and variances, a portfolio's expected return and variance can be calculated.

II. Portfolio Expected Return

The expected return of a portfolio is the **weighted average** of the individual assets' expected returns:

$$E(R_p) = \sum_{i=1}^{n} w_i E(R_i)$$

where:

- w_i = portfolio weight of asset i
- $E(R_i)$ = expected return of asset i
- $\sum_{i=1}^{n} w_i = 1$

III. Case Study: Three-Stock Portfolio

Consider three stocks (A, B, and C) under two economic states:

State	Probability	Stock A	Stock B	Stock C
Boom	0.4	10%	15%	20%
Bust	0.6	8%	4%	0%

Method 1: Individual Expected Returns First

Calculate the expected return for each stock:

$$E(R_A) = (0.4 \times 10\%) + (0.6 \times 8\%) = 4\% + 4.8\% = 8.8\%$$
$$E(R_B) = (0.4 \times 15\%) + (0.6 \times 4\%) = 6\% + 2.4\% = 8.4\%$$
$$E(R_C) = (0.4 \times 20\%) + (0.6 \times 0\%) = 8\% + 0\% = 8\%$$

With portfolio weights $w_A = 50\%$, $w_B = 25\%$, $w_C = 25\%$:

$$E(R_p) = 0.5(8.8\%) + 0.25(8.4\%) + 0.25(8\%) = 4.4\% + 2.1\% + 2\% = 8.5\%$$

Method 2: State-by-State Portfolio Returns First

First calculate the portfolio return in each economic state:

$$R_p(\text{Boom}) = 0.5(10\%) + 0.25(15\%) + 0.25(20\%) = 5\% + 3.75\% + 5\% = 13.75\%$$
$$R_p(\text{Bust}) = 0.5(8\%) + 0.25(4\%) + 0.25(0\%) = 4\% + 1\% + 0\% = 5\%$$

Then calculate the portfolio expected return:

$$E(R_p) = 0.4(13.75\%) + 0.6(5\%) = 5.5\% + 3\% = 8.5\%$$

Key Insight

Method 2 (state-by-state) is particularly useful for calculating portfolio variance because it provides the full distribution of possible portfolio returns.

IV. Portfolio Variance and Diversification

Critical Concept

Portfolio variance is NOT a simple weighted average of individual asset variances. The actual portfolio standard deviation is often lower than the weighted average of individual standard deviations.

Example: Diversification Benefit

Even if the weighted average of individual standard deviations suggests 27.5%, the actual calculated portfolio standard deviation may be only 17.5%—a reduction of 10 percentage points due to diversification.

Three-Stock Portfolio Variance Calculation

Using the same three stocks with $w_A = 0.5$, $w_B = 0.25$, $w_C = 0.25$:

$$
\begin{aligned}
E(R_p) &= 8.5\% = 0.085 \\
\sigma_p^2 &= 0.4(0.1375 - 0.085)^2 + 0.6(0.05 - 0.085)^2 \\
&= 0.4(0.0525)^2 + 0.6(-0.035)^2 \\
&= 0.4(0.00275625) + 0.6(0.001225) \\
&= 0.0011025 + 0.000735 = 0.0018375 \\
\sigma_p &= \sqrt{0.0018375} = 0.04287 = 4.29\%
\end{aligned}
$$

Diversification Benefit

By holding a variety of assets, investors achieve a **diversification benefit** that reduces the portfolio's overall risk.

V. Understanding Risk and "Surprises"

Risk is defined as the **deviation from expected outcomes**. Actual market returns often differ from expected returns due to **surprises** or **innovations**.

Total Return Decomposition

$$\text{Total Return} = \text{Expected Return} + \text{Unexpected Return}$$

Example: GDP Announcement

- Shareholders forecast GDP growth: 5%
- Government announces actual growth: 1.5%
- **Surprise:** -3.5% (the unexpected component)
- This surprise affects the total return

These surprises can be positive or negative and are the source of investment risk.

VI. Systematic vs. Unsystematic Risk

Risk is categorized into two primary types:

Systematic Risk (Market Risk)

- Affects the **entire market** or a large segment of it
- Sources: recessions, inflation, interest rate changes, political events (wars), natural disasters
- **Cannot be eliminated through diversification**
- Always present in any investment

Unsystematic Risk (Unique/Firm-Specific Risk)

- Associated with a **specific company or industry**
- Sources: product failure, labor strikes, change in leadership, supply chain issues
- **Can be reduced or eliminated through portfolio diversification**

Visual Representation

$$\text{Total Risk} = \text{Systematic Risk} + \text{Unsystematic Risk}$$

As the number of assets in a portfolio increases:

- Unsystematic risk approaches zero
- Systematic risk remains constant

VII. Strategies for Diversification

To minimize unsystematic risk, investors should use **portfolio diversification**:

Diversification Strategies

1. **Across Asset Classes:** Combine stocks, bonds, real estate, and commodities
2. **Across Industries:** Invest in technology, healthcare, consumer goods, financial services
3. **Geographic Diversification:** Invest in different countries and regions
4. **Investment Styles:** Combine growth stocks with value stocks
5. **Use Index Funds/ETFs:** Exchange Traded Funds automatically provide broad diversification

Example: Growth vs. Value Stocks

- **Growth Stocks:** High-risk technology companies (potential for high returns)
- **Value Stocks:** Stable, established businesses (e.g., Nestle, consumer staples)
- Combining both reduces overall portfolio volatility

Key Conclusion

While unsystematic risk can be minimized through diversification, **systematic risk will always remain**. Investors cannot eliminate market-wide risks.

VIII. Course Logistics

Midterm Syllabus

- Chapter 5: Time Value of Money (Amortization Table)
- Chapter 6: Interest Rates
- Chapter 7: Bonds and Their Valuation
- Chapter 13: Risk and Return (current topic)

Preparation

Students should practice calculating:

- Expected returns for individual assets and portfolios
- Variance and standard deviation
- Portfolio variance using state-by-state method

Summary of Key Formulas

$$E(R_p) = \sum_{i=1}^{n} w_i E(R_i)$$

$$\sigma_p^2 = \sum_{i=1}^{n} p_i (R_{p,i} - E(R_p))^2$$

$$\text{Total Return} = E(R) + \text{Unexpected Return}$$

Risk Type	Elimination Through Diversification
Systematic Risk	Cannot be eliminated
Unsystematic Risk	Can be eliminated